Pre-class progress check

By now, you should have

1. Started thinking about your group project proposals (due 11.02 by 11.59pm)
2. Reviewed slides and code from classes 1-6

1. Reviewed slides and code from class 7
2. Reviewed graded assignment 1 + answer key

1. Started working on group project
2. Reviewed slides from class 8 (today)

University of Pittsburgh | School of Public Health

Unsupervised learning (I): Clustering (I)
- k-means
- hierarchical clustering

BIOST 2155

Class 8
October 24th, 2025

On clustering

Can we discover subgroups among the observations or variables?

When we cluster the observations of a data set, we want to partition them into distinct groups so that the observations within each group are quite similar to each other, while observations in different groups are quite different from each other.

Market Segmentation: Finding groups of customers with similar purchasing behavior.

Biology: Grouping genes with similar expression patterns, or patients by disease subtype.

Image Recognition: Grouping similar images together.

Different approaches:

1. **K-means clustering:** partition the observations into a pre-specified number of clusters.
2. **Hierarchical clustering:** do not know in advance how many clusters we want. Get a tree-like visual representation of the $n-$many observations, called a dendrogram, that allows us to view at once the clusterings obtained for dendrogram each possible number of clusters, from 1 to n.

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fig1
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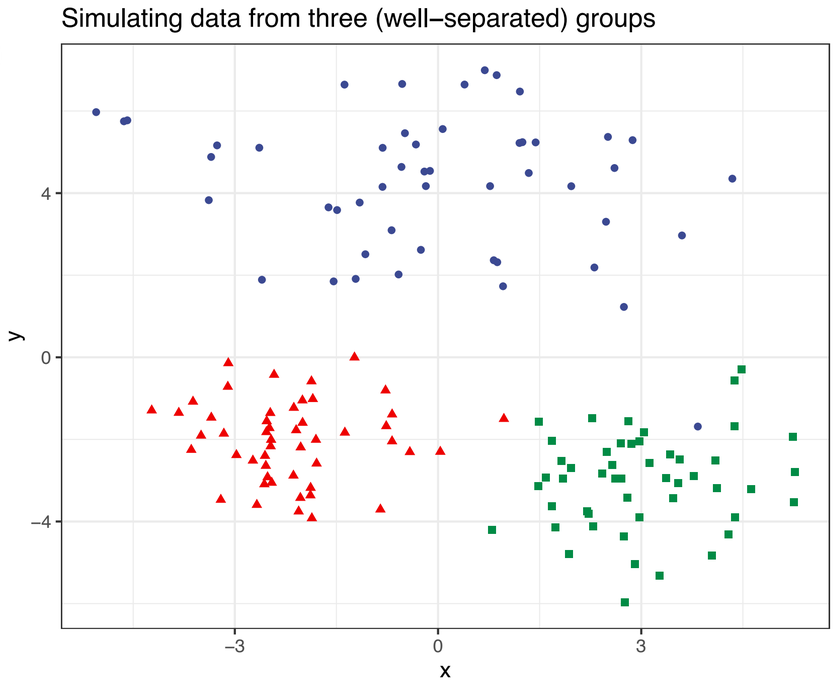
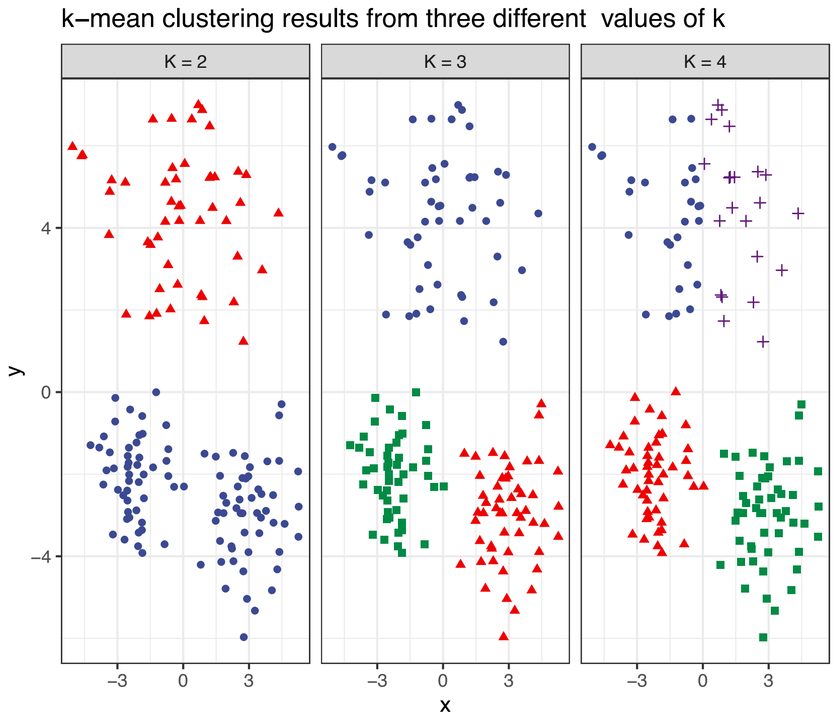
K-means clustering

Overview

1. Simple and elegant approach for partitioning a data set into K distinct, non-overlapping clusters.
2. To perform K-means clustering, we must first specify the desired number of clusters K
3. Then the K-means algorithm will assign each observation to <u>exactly</u> one of the K clusters.
4. Good clustering = within-cluster variation is <u>as small as possible</u>.





K-means clustering

Defining notation

1. Let $C_1, C_2, \ldots, C_K$ denote sets containing the indices of the observations in each cluster.
2. Each observation belongs to at least one of the K clusters

$$\cup_{i=1}^{K} C_i = \{1,2,\ldots,n\}$$

3. No observation belongs to more than one cluster

$$C_k \cap C_l = \emptyset \text{ for } 1 \leq k \neq l \leq K$$

$i-$th observation is in the $l-$th cluster: $i \in C_l$

Good clustering: within-cluster variation is as small as possible.

Within-cluster variation for cluster C_l = $W(C_l)$

1	2	3	4	5	6	7	8	9	10
1	2	**3**	**4**	**5**	*6*	*7*	*8*	9	10

K-means clustering

Clustering is an optimization problem

1. The within-cluster variation for cluster C_k: $W(C_k)$ - the amount by which the observations within a cluster differ from each other.
2. "Good clustering" = solving the problem

$$\min_{C_1,\ldots,C_K} \left\{ \sum_{k=1}^{K} W(C_k) \right\}$$

3. Many different ways to define $W(C_k)$....

$$W(C_k) = \frac{1}{n(C_k)} \sum_{i,j \in C_k} \| \boldsymbol{x}_i - \boldsymbol{x}_j \|_0^0$$

$$W(C_k) = \frac{1}{n(C_k)} \sum_{i,j \in C_k} \| \boldsymbol{x}_i - \boldsymbol{x}_j \|_1^1$$

$$W(C_k) = \frac{1}{n(C_k)} \sum_{i,j \in C_k} \| \boldsymbol{x}_i - \boldsymbol{x}_j \|_2^2$$

The general $\mathscr{L}_p$ norm for a vector $\boldsymbol{x} = (x_1, x_2, \ldots x_n)^T$ is $\|\boldsymbol{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$

Optimizing for $\mathscr{L}_2$ norm

Lloyd's algorithm

- Our optimization problem now is:

$$\min_{C_1,\ldots,C_K} \left\{ \sum_{k=1}^{K} \frac{1}{n(C_k)} \sum_{j,j' \in C_k} \| \boldsymbol{x}_j - \boldsymbol{x}_{j'} \|_2^2 \right\}$$

- **Step 1:** randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.
- **Step 2:** iterate until the cluster assignments stop changing
  - For each of the K clusters, compute the cluster centroid. The k-th cluster centroid is the vector of the p feature means for the observations in the k -th cluster.
  - Assign each observation to the cluster whose centroid is closest (where **closest** is defined using $\mathscr{L}_2$ norm)

What is a good K?

This algorithm is guaranteed to decrease the value of the objective

It is not guaranteed to give the global minimum

Expect an assignment problem on this….

How would the algorithm change for other penalties?

(doable for $\mathscr{L}_1$ or $\mathscr{L}_\infty$)
(not an assignment problem)

Hierarchical clustering

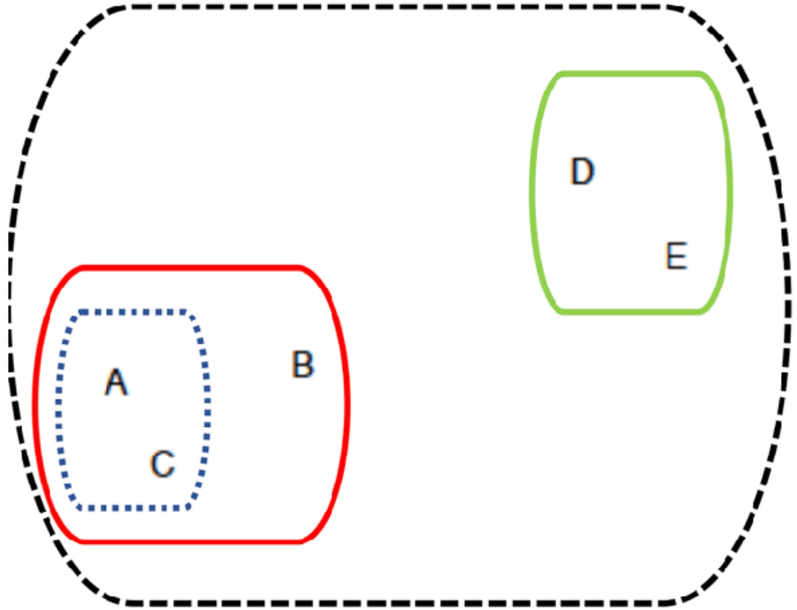
If you can't commit to K just yet…

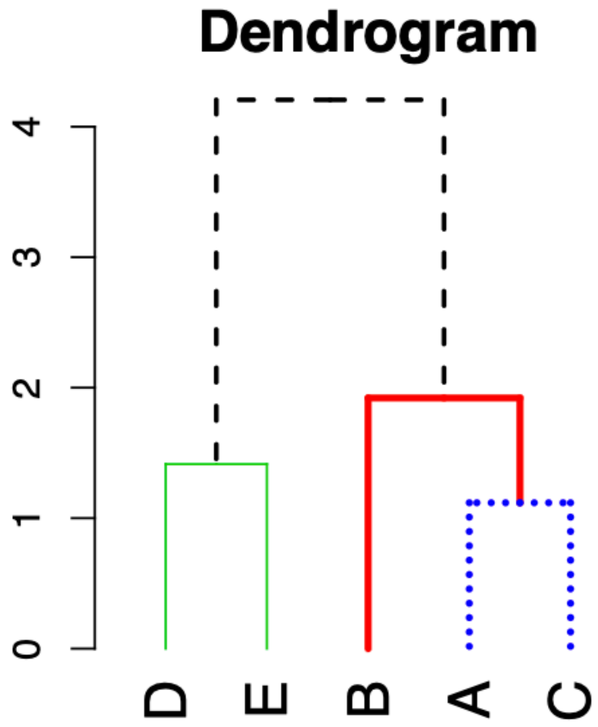
- K-means clustering requires us to pre-specify the number of clusters K. This can be a disadvantage (later we discuss strategies for choosing K).

- Hierarchical clustering is an alternative approach which does not require that we commit to a particular choice of K.
  - Bottom-up or agglomerative clustering.
  - Common type of hierarchical clustering,
  - Dendrogram is built starting from the leaves and combining clusters up to the trunk.

Start with each point in its own cluster.
- Identify the closest two clusters and merge them.
- Repeat.

End when all points are in a single cluster.





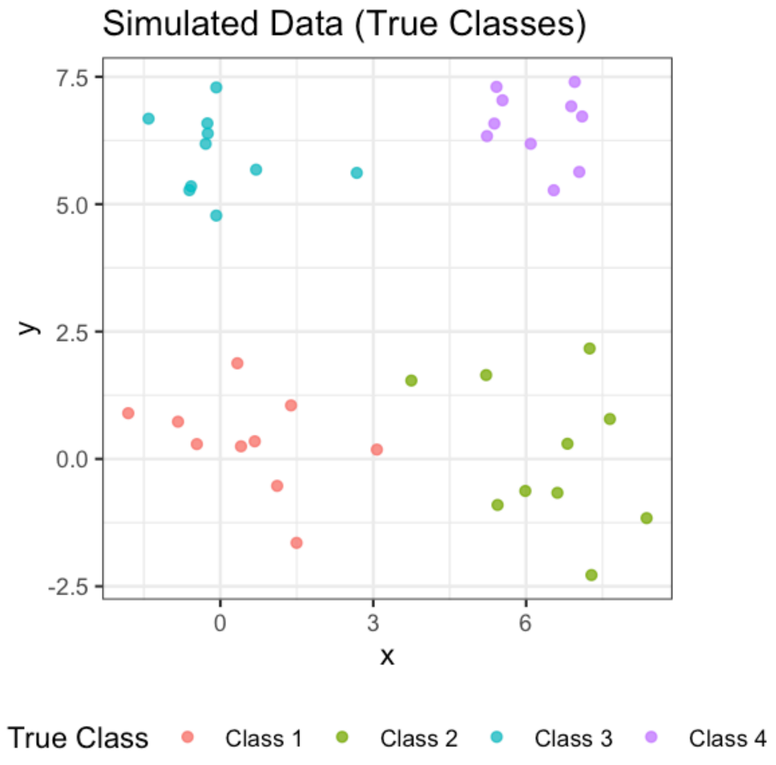
Hierarchical clustering

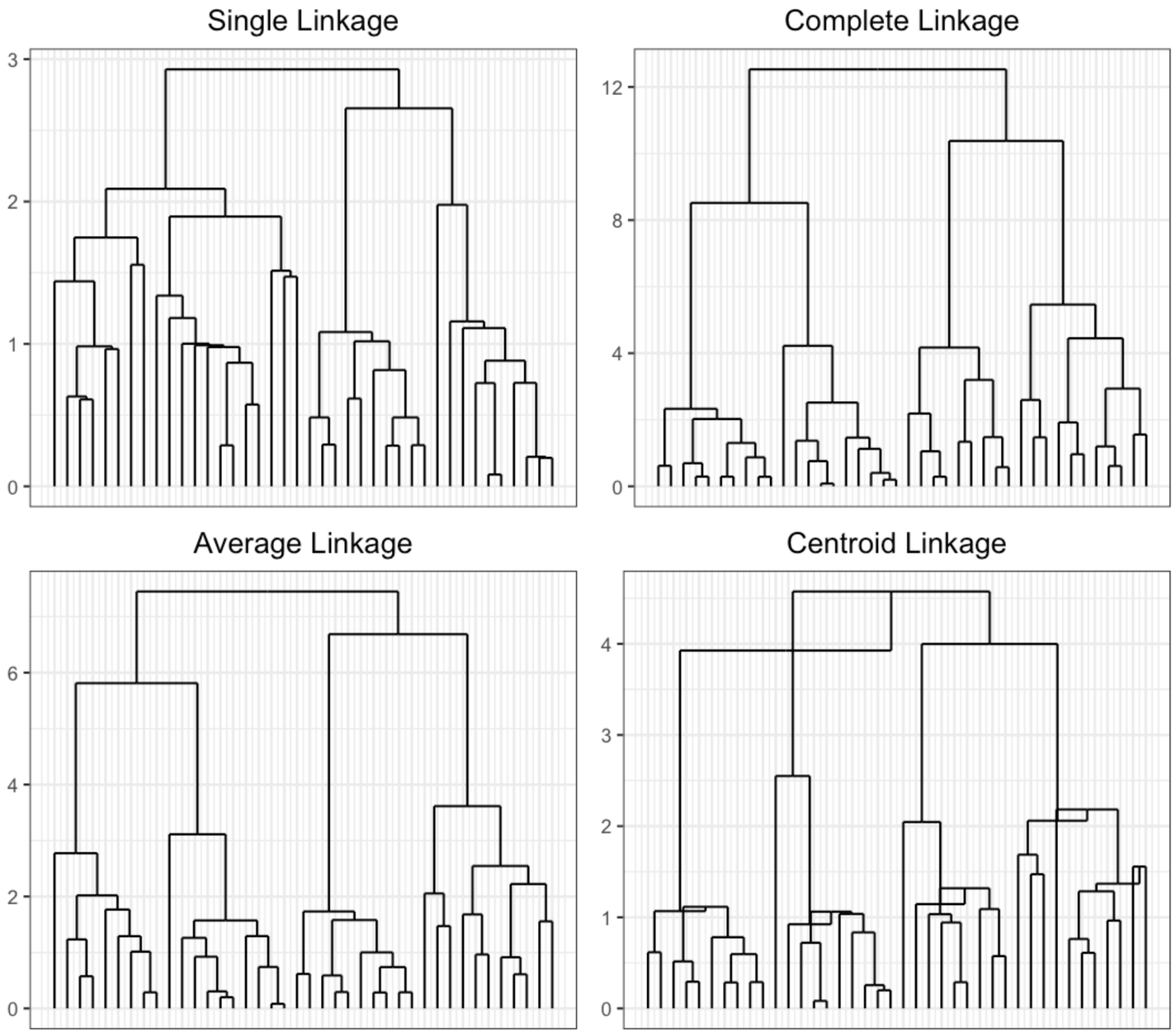
If you can't commit to K just yet…

- Begin with n observations and measure (using $\mathscr{L}_p$ norm) all $n(n-1)/2$ **inter-observation distances**. Treat each observation as it's own cluster.
- For $i = n, n-1, \ldots, 2$:
  - Examine all pairwise **inter-cluster dissimilarities** among the i clusters and identify the pair of clusters that are **most similar** and fuse them.
  - Dissimilarity between two clusters indicates the height at which in the dendrogram at which the fusion is placed.
  - Compute new pairwise **inter-cluster dissimilarities** among the $i-1$ remaining clusters. Repeat step above.

Dissimilarity type	Description
Complete	Maximal intercluster dissimilarity. Compute all pairwise **distances** between the observations in cluster A and the observations in cluster B, and record the largest of these **distances**.
Single	Minimal intercluster dissimilarity. Compute all pairwise **distances** between the observations in cluster A and the observations in cluster B, and record the smallest of these **distances**.
Average	Mean intercluster dissimilarity. Compute all pairwise **distances** between the observations in cluster A and the observations in cluster B, and record the average of these **distances**.
Centroid	**Distance** between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B.

fig2


Simulated Data (True Classes)
7.5
5.0
2.5
0.0
-2.5
y
0
3
6
x
True Class
Class 1
Class 2
Class 3
Class 4


Single Linkage
3
2
1
0
Complete Linkage
12
8
4
0
Average Linkage
6
4
2
0
Centroid Linkage
4
3
2
1
0

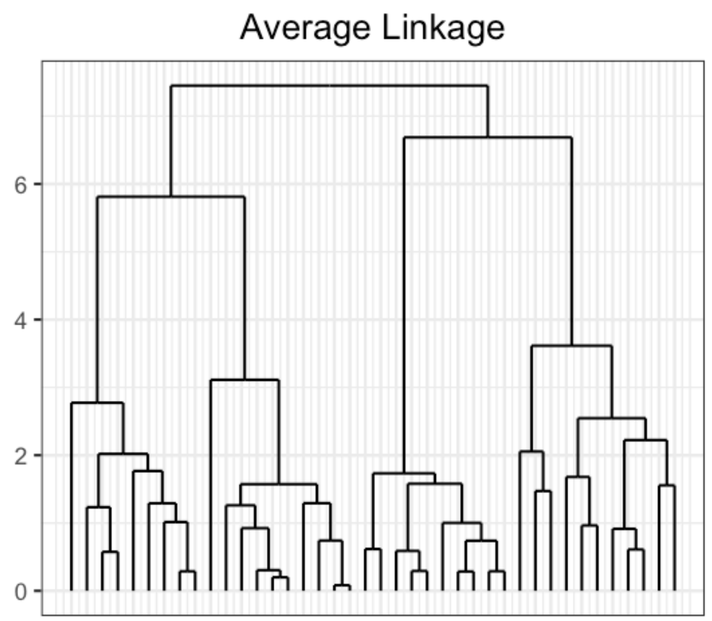
Hierarchical clustering

Using dendrograms to commit to K

- Look for the Longest Vertical Line: On your dendrogram look for the longest vertical line that doesn't get "crossed" by any other cluster merges (horizontal lines).
- Draw a Horizontal Line: Imagine drawing a horizontal line that cuts right through the middle of that longest vertical segment.
- Count the Intersections: Count how many vertical lines your new horizontal line crosses.

Scaling matters! Always center and scale!!

- The Y-axis on the dendrogram represents the distance or "dissimilarity" between the clusters being merged.
- A short vertical line means two clusters that were already "pretty close" got merged.
- A long vertical line represents a "big jump." It means the algorithm had to merge two clusters that were very far apart (i.e., very dissimilar).
- By drawing your horizontal "cut line" below that big jump, we're saying, "not willing to accept any merges between clusters that are **this** different."



The Gap Statistic

A structured way of choosing K…

Gap Statistic measures total within-cluster sum of squares:

$$W_K = \sum_{k=1}^{K} \sum_{i \in C_k} \| \boldsymbol{x}_i - \boldsymbol{\mu}_k \|_2^2,$$ where $\boldsymbol{\mu}_k$ is the mean vector for C_k

We compute $\log(W_K)$ from **real data** and compare with $\mathbb{E}^\star\left(\log(W_K)\right)$, which measures behavior of W_K under '**null reference data**'

For $k = 1,2,\ldots,K_{\text{max}}$:

- Cluster observed data $\mathscr{D}$ with k clusters and compute $\log(W_k)$.
- Generate B "null" reference datasets by sampling uniformly from the bounding box of $\mathscr{D}$: $\mathscr{D}_1^\star, \mathscr{D}_2^\star, \ldots, \mathscr{D}_B^\star$
  - Get $\log(W_{k,b}^\star)$, $1 \leq b \leq B$
  - Estimate $\hat{\mathbb{E}}^\star\left(\log(W_k)\right) = B^{-1} \sum_{b=1}^{B} \log(W_{k,b}^\star)$
  - $\text{Gap}_k = \hat{\mathbb{E}}^\star\left(\log(W_k)\right) - \log(W_k)$
- Choose value of k that maximizes Gap_k

The crux of the gap statistic and **permutation tests** are very similar: you set up a "contest" between two datasets:

- Observed data
- Simulated null data: create a fake dataset with no clusters (uniform random sample). Run the exact same k-means algorithm on this fake data. This gives you a "baseline clustering" score.

Post-class debrief

Today we discussed…

The fundamental goal of clustering is to discover subgroups within data. We want to partition observations into distinct groups so that items within the same group are very similar to each other, while items in different groups are very dissimilar. This is a core technique in unsupervised learning and has many applications, such as market segmentation, grouping genes with similar expression patterns, and organizing images.

- **K-Means Clustering:** This method partitions the dataset into a pre-specified number of K distinct, non-overlapping clusters by trying to make the within-cluster variation as small as possible.
  - Advantages: It is a simple and elegant approach to partitioning a dataset.
  - Disadvantages: Must pre-specify the number of clusters, K, which is often unknown. Further, the algorithm is not guaranteed to find the single best (global minimum) solution and can get stuck in a sub-optimal local minimum.

- **Hierarchical Clustering:** This method does not require specification of the number of clusters in advance; instead, it builds a tree-like visual representation called a dendrogram that shows the cluster merges at every level.
  - Advantages: Its main advantage is that it does not require pre-specified K. It produces a dendrogram, which is a rich visualization that depicts see the clustering structure at all levels and can be used to select a good K.
  - Disadvantages: The choice of "linkage" (e.g., complete, single, average) can significantly affect the results. The algorithm is more computationally intensive, as it must compute all pairwise distances between observations.

- The Gap Statistic is a formal, structured method for choosing the optimal number of clusters, K, for an algorithm like k-means.
- Intuition: It answers the question, "How much better is the clustering than the clustering we get on 'null' data with no structure?"

We are ending a module!! Assignment incoming…

Assignment based on material from classes 5 - 8

Assignment 2 will be harder than Assignment 1

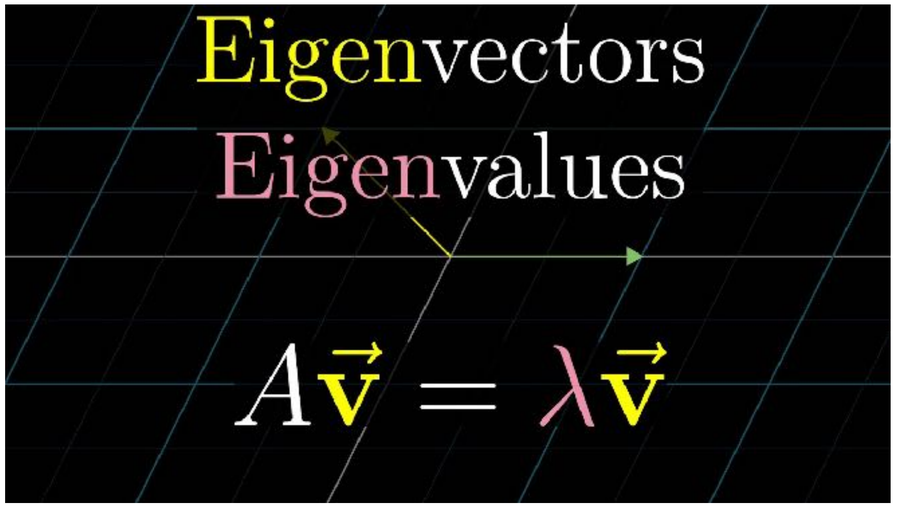
Three sections

1. Theory (two questions for <u>all</u>)
2. Application A: Classification
3. Application B: Clustering

Released today after class, due by 11:59PM on Sunday 2025-11-09.

We will discuss principal component analysis in the next class - to refresh your linear algebra knowledge please read pages 98-119 from Mathematics for Machine Learning (linked on Canvas).

I will assume you know eigendecomposition.



That's all Folks!